WORKED SOLUTION

STEP I 2016, Question 1

Pure • Polynomial Identities and Factorisation

STEP I 2016, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s1-q1

The key idea

The quotient $q_n(x)$ is really a polynomial: $x+1$ divides $x^{2n+1}+1$ because $2n+1$ is odd, and the quotient is an alternating geometric sum. So part (i) is a matter of expanding p_n and comparing coefficients. The pay-off in part (ii) comes from the shape of p_n: when $2n+1$ times x is a perfect square, p_n is a difference of two squares, and so it factorises.

Solution

Part (i)

For $x \neq -1$, summing the geometric series with first term 1 and ratio $-x$ gives

$$1 - x + x^2 - \cdots + x^{2n} = \sum_{k=0}^{2n} (-x)^k = \frac{1-(-x)^{2n+1}}{1+x} = \frac{1+x^{2n+1}}{1+x},$$

using $(-x)^{2n+1} = -x^{2n+1}$ as $2n+1$ is odd. So

$$q_n(x) = x^{2n} - x^{2n-1} + x^{2n-2} - \cdots - x + 1,$$

with all $2n+1$ coefficients equal to ± 1, alternating in sign. We now expand p_n in each case.

$n = 1$.

$$p_1(x) = (x+1)^2 - 3x = x^2 + 2x + 1 - 3x = x^2 - x + 1 = q_1(x).$$

$n = 2$.

$$\begin{aligned} p_2(x) &= (x+1)^4 - 5x(x^2+x+1) \\ &= (x^4 + 4x^3 + 6x^2 + 4x + 1) - (5x^3 + 5x^2 + 5x) \\ &= x^4 - x^3 + x^2 - x + 1 = q_2(x). \end{aligned}$$

$n = 3$. First $(x^2+x+1)^2 = x^4+2x^3+3x^2+2x+1$, so

$$\begin{aligned} p_3(x) &= (x+1)^6 - 7x(x^2+x+1)^2 \\ &= (x^6+6x^5+15x^4+20x^3+15x^2+6x+1) - (7x^5+14x^4+21x^3+14x^2+7x) \\ &= x^6 - x^5 + x^4 - x^3 + x^2 - x + 1 = q_3(x). \end{aligned}$$

So $p_n(x) \equiv q_n(x)$ for $x \neq -1$ when $n = 1, 2, 3$, as required.

$n = 4$. One value of x is enough to break an identity. At $x = 1$,

$$p_4(1) = 2^8 - 9 \cdot 1 \cdot 3^3 = 256 - 243 = 13, \qquad q_4(1) = \frac{1+1}{1+1} = 1,$$

so $p_4 \not\equiv q_4$. (Expanding fully gives $p_4(x) = x^8 - x^7 + x^6 + 2x^5 + 7x^4 + 2x^3 + x^2 - x + 1$, which goes wrong from the x^5 term onwards.)

Part (ii)

Ask yourself Which n and which x should we use, and how can p_n possibly factorise?

Matching $\dfrac{x^{2n+1}+1}{x+1}$ to the given numbers fixes x and n. The form of p_n then helps: $(x+1)^{2n} = ((x+1)^n)^2$ is always a square, and $(x^2+x+1)^{n-1}$ is a square when n is odd. So if $(2n+1)x$ is also a perfect square, $p_n(x)$ is a difference of two squares.

(a) Take $n = 1$ and $x = 300$, so $2n+1 = 3$ and $3x = 900 = 30^2$. By part (i),

$$\begin{aligned} \frac{300^3+1}{301} = q_1(300) = p_1(300) &= 301^2 - 3 \cdot 300 \\ &= 301^2 - 30^2 \\ &= (301-30)(301+30) = 271 \times 331. \end{aligned}$$

Neither factor is 1. (Directly, $300^2 - 300 + 1 = 89\,701 = 271 \times 331$.)

(b) Since $7^{49} = (7^7)^7$, take $n = 3$ (so $2n+1 = 7$) and $x = 7^7$. Then $(2n+1)x = 7 \cdot 7^7 = 7^8 = (7^4)^2$, a perfect square, and $n - 1 = 2$ is even. By part (i),

$$\begin{aligned} \frac{7^{49}+1}{7^7+1} = q_3(7^7) = p_3(7^7) &= (7^7+1)^6 - 7^8\,(7^{14}+7^7+1)^2 \\ &= [(7^7+1)^3]^2 - [7^4(7^{14}+7^7+1)]^2, \end{aligned}$$

so

$$\frac{7^{49}+1}{7^7+1} = \Big[(7^7+1)^3 - 7^4(7^{14}+7^7+1)\Big]\Big[(7^7+1)^3 + 7^4(7^{14}+7^7+1)\Big].$$

The second factor is clearly greater than 1. For the first, $(7^7+1)^3 > 7^{21}$ while

$$7^4(7^{14}+7^7+1) = 7^{18} + 7^{11} + 7^4 < 7^{19},$$

so the first factor exceeds $7^{21} - 7^{19} = 48 \cdot 7^{19} > 1$. Neither factor is 1, and we have the required factorisation. (Multiplying out the brackets, the factors could also be written $7^{21} + 3 \cdot 7^{14} + 3 \cdot 7^7 + 1 \mp (7^{18} + 7^{11} + 7^4)$.)

Checks

- Put $x = 1$ in part (i): $p_1(1) = 4 - 3 = 1$, $p_2(1) = 16 - 15 = 1$, $p_3(1) = 64 - 63 = 1$, all equal to $q_n(1) = 1$, while $p_4(1) = 13$.
- In (a), $271 \times 331 = 89\,701 = 300^2 - 300 + 1$.
- In (b), a computer check of the exact integers confirms that the product of the two factors equals $(7^{49} + 1)/(7^7 + 1)$; the factors are about 5.57×10^{17} and 5.60×10^{17}.

Common slips

- Forgetting to say why neither factor is 1; in (b) this needs a short size estimate.
- In (b), using $n = 24$ or $x = 7$: the exponent 49 must be written as 7×7, giving $x = 7^7$ and $2n + 1 = 7$.

Beyond STEP

The trick here is a special case of Aurifeuillean factorisation: identities such as $x^4 + 4y^4 = (x^2 + 2y^2 - 2xy)(x^2 + 2y^2 + 2xy)$, which split numbers of the form $a^k \pm 1$ further than the obvious algebraic factors would suggest. They come from cyclotomic polynomials, which appear in first-year algebra and number theory courses, and they were used in the hunt to factor large numbers of the form $2^k \pm 1$. Davenport's *The Higher Arithmetic* is a good place to read more about factorisation and primes.

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